World Trivia



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Professor Swettlen, Professor Beach, ECEN-1400 – Fall 2025

Engineering
UNIVERSITY OF COLORADO BOULDER

Project Background

Mission Statement:

Our goal is to create an accessible and enjoyable trivia game for the elderly or those with mobility limitations to reminisce over past travel while actively engaging in mental stimulation.

Game Rules

- 4 Players
- The player with the most correctly answered questions out of 5 wins the game
- The 1st player will navigate to a continent using the joystick
- All players will have 6 seconds to answer the question presented
- Each correctly answered question earns the player a point
- Next player will use the joystick to choose one of the 7 continents and the game continues until all 5 questions have been asked

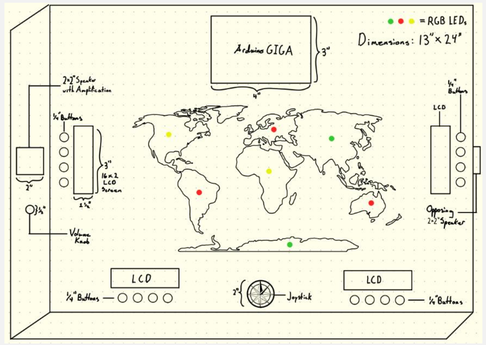
Challenges Faced

- Soldering joints were too delicate, so had to find better way to attach components, settling on wire nuts
- Backpacks on two of the small LCD Screens were damaged, so had to replace them
- Larger Laser cutter was out for repair, so had to redesign board for smaller laser cutter



Engineering Design & Process

The electrical design communicates from our main controller, Arduino GIGA to our LCD screens, LEDs and joystick. These elements each require a feed of 5 volts, so we supply with a 7.4V rechargeable lithium-ion battery.





Dimensions

- Overall Footprint: 24" X 21"
- Height: 3"

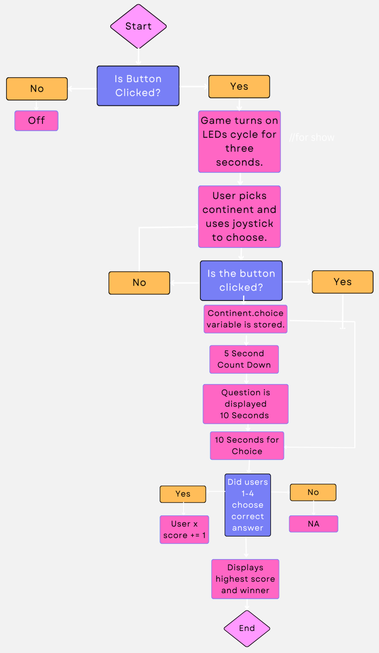
Electrical Elements

- Arduino Giga
- Arduino Shield LCD Screen
- 7.3 Volt Lithium-Ion Battery Power Supply
- 7 LEDs
- 16 Push Buttons
- Four 2x16 LCD Screens

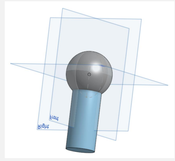
Software Components

- Joystick correlates to an LED, when pressed a question is displayed on Arduino Shield
- Buttons correlate to an answer which can be chosen by each player
- Smaller LCD screens display points

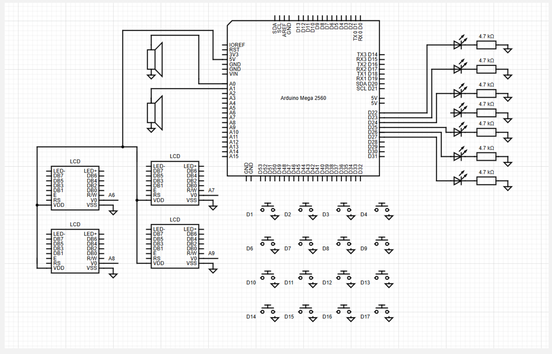
Software Flow Charts



Main Code – Block Diagram



For the housing of our game, we have laser engraved the wood game board and built interior housings for our internal parts.



Electrical Schematic

Testing

Knowing our biggest rick factor was the LCD screens and our Arduino GIGA, we put the most effort into testing those elements

- Ensured elements worked individually
- Components were tested on a breadboard and separate Arduino before soldering
- Troubleshooted hardware with all connections in place

Our next biggest concern was the coding, which was much more difficult to test and troubleshoot.

- We ensured the joystick navigated to the correlating continent and LED when moved
- Checked that buttons provided analog response within coding
- Checked random question generation within code
- Had a test group play the game and provide feedback

Prototyping

Initial Processes:

- Went through several phases of game board dimensions, settling on 24”x21”
- The joystick was initially too big, so required scaling down

Lessons Learned

Technical Development: Members learned OnShape, laser machining, Arduino coding, soldering, and 3D design to help pull the project together and ensure its success.

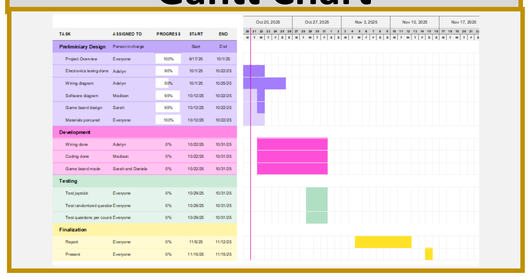
Future Developments

If we had more time, as a group we would add in the originally planned speakers and add to the visual design.

Budget

Quantity & Product	Price per
Arduino GIGA with LCD Screen	$121.00
2x16 LCD Screens	$15.00
RGB LEDs	$16.00
Joystick	$7.00
Speaker Wires	$6.00
Acrylic Sheet	$33.00
Wood	$30.00
Power Supply	$29.00
Total	$257.00

Gantt Chart



Acknowledgments

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